

# Operating Systems Exam 1

個人成績報告

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滿分: **100 / 101** · 正確率: **98%**

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## 題目詳情

Q1 簡答題 (2分)

✓ 正確 2 / 2

How many times does the following code print "hello"?

```
main() {  
    if (fork() == 0) {  
        fork(); printf("hello\n");  
    };  
    printf("hello\n");  
}
```

你的答案

5次

參考答案

5 (The child forks again inside the IF; the parent only prints once).

## Q2 簡答題 (2分)

✓ 正確 2 / 2

What is the result of this code?

```
main() {  
    fork();  
    execl("/bin/echo", "echo", "hello", 0);  
    printf("done\n");  
}
```

你的答案

hellohello

參考答案

"hello" is printed twice; "done" is never printed. (execl replaces the process image).

## Q3 問答題 (2分)

△ 部分 1 / 2

List two advantages of kernel-level threads over user-level threads.

你的答案

Responsiveness:執行中process內部的threads被block住，CPU仍可以切換給其他available threads執行，故process仍可繼續執行

Resource sharing:process內的多個threads共享code section，所以多個工作可以在同一個memory space下同時工作

參考答案

- a. Multiprocessor utilization (can run on different CPUs).
- b. One thread blocking doesn't block the entire process.

TA 評語

第2點 resource sharing是所有threads的通性，並非kernel-level threads特有優點。

## Q4 問答題 (2分)

✓ 正確 2 / 2

What is busy waiting (spinlock)?

你的答案

process/thread要存取某個共享資源或lock時，不block/不sleep，在CPU上loop continuous checking直到達成某些條件才繼續執行。例:process要求能否進入critical section或polling i/o device是否ready。

參考答案

Busy waiting, also known as a spin lock, is the situation when a thread is looping, continuously checking whether a condition exists that will allow it to continue. It is often used to check if a thread can enter a critical section or to poll a device to see if data is ready (or written/sent).

Q5 問答題 (2分)

✓ 正確 2 / 2

Describe the concept of "demand paging" as a virtual memory implementation. Why is it commonly used?

你的答案

demand paging是指要用到的page才載入memory。可以更有效的使用memory空間

參考答案

- a. Demand paging loads pages into memory only when they are needed (on page fault), not at process start.
- b. It is commonly used because it reduces initial load time, saves memory by not loading unused pages, and allows larger programs than physical memory.

Q6 問答題 (3分)

✓ 正確 3 / 3

What is the difference between a preemptive scheduler and a non-preemptive scheduler? Provide an example scenario where preemptive scheduling is beneficial.

你的答案

preemptive scheduler:可以搶奪其他process的資源(即使對方不是自願放棄)

non-preemptive scheduler:除非process自願放棄resource，否則不能被搶走

當有一個job非常緊急的時候，需要優先執行，preemptive scheduler就能讓他馬上取得可完成工作之資源去完成工作

參考答案

- a. Preemptive scheduling can interrupt a running thread before its burst ends to switch to another.
- b. Non-preemptive runs a thread to completion or voluntary yield.
- c. Preemptive is beneficial in time-sharing systems for fair CPU sharing among interactive users, preventing long-running tasks from monopolizing the CPU.

Q7 問答題 (2分)

✓ 正確 2 / 2

Explain Belady's Anomaly in the context of page replacement algorithms. Which replacement algorithm has this anomaly?

你的答案

常發生在FIFO，是一個反常現象，給更多的page frames數量，page fault ratio不降反升。

參考答案

- a. Belady's Anomaly occurs when increasing the number of page frames leads to more page faults.
- b. The FIFO replacement has Belady's Anomaly. It occurs because FIFO doesn't consider access patterns; older pages might be replaced despite future use.

**Q8 問答題** (3分)

✓ 正確 3 / 3

What is priority inversion? Describe a scenario where busy waiting (spinlock) can lead to it. How can priority inheritance solve this issue?

**你的答案**

一個low priority job A已經進入critical section並取得lock，此時有一個high priority job B也想進入critical section但沒有lock，B會busy waiting(spinlock)，由於是priority scheduling，scheduler會把CPU不斷給B，導致low priority job A拿不到CPU去完成工作並release lock，形成優先權顛倒的現象。  
解決的方法就是priority inheritance，把low priority job A的優先權提升到所有需要此共享資源中job裡面的最高，等他拿到CPU完成工作再調回原本的優先權

**參考答案**

- Priority inversion occurs when a higher-priority task is blocked by a lower-priority one.
- Scenario: Low-priority A holds a lock; high-priority B spins waiting. The scheduler runs B over A, preventing A from releasing the lock.
- Priority inheritance solves this by temporarily boosting the priority of the lock holder (A) to the level of the waiting high-priority task (B).

**Q9 簡答題** (2分)

✓ 正確 2 / 2

In a system with a TLB hit ratio of 85%, if TLB access time is 15ns and memory access time is 100ns, calculate the effective memory access time (EAT).

**你的答案**

130ns

**參考答案**

$EAT = 0.85 * (15 + 100) + 0.15 * (15 + 100 + 100) = 0.85 * 115 + 0.15 * 215 = 97.75 + 32.25 = 130ns.$

**Q10 問答題** (2分)

✓ 正確 2 / 2

Given a disk with 150 tracks (0-149), head at track 80 moving towards increasing numbers. Requests are in tracks of 20, 45, 10, 120, 130, 5, 90. What is the service order using SCAN and C-SCAN algorithms?

**你的答案**

先sort:5,10,20,45,90,120,130  
SCAN:90,120,130,45,20,10,5  
C-SCAN:90,120,130,5,10,20,45

**參考答案**

- SCAN: 90, 120, 130, 45, 20, 10, 5. (Moves to the end, then reverses).
- C-SCAN: 90, 120, 130, 5, 10, 20, 45. (Moves to the end, jumps to the start, then continues).

Q11 問答題 (6分)

✓ 正確 6 / 6

List three attributes that are shared among threads in the same process, and three that are not shared.

你的答案

share:code section、data section、global 變數  
not share:thread id,program counter,stack

參考答案

- a. Shared: Memory space, file descriptors, global variables.
- b. Not shared: Thread ID, registers, stack.

Q12 問答題 (3分)

✓ 正確 3 / 3

Briefly describe the concepts of "thread," "process," and "program." How do they relate?

你的答案

program:passive entity，有code section和static data。  
process:program in execution。  
thread:process的execution flow，thread間共享memory space和其他resources，不共享program counter、cpu register、stack。

參考答案

- a. A program is a static file containing code.
- b. A process is an active execution instance of that program.
- c. A thread is a unit of execution within a process. A process can contain multiple threads that share the same resources.

Q13 簡答題 (2分)

✓ 正確 2 / 2

In a UNIX inode with 12 direct blocks and 1 single indirect block, assuming a block size of 8KB and address size of 4 bytes, calculate the maximum file size.

你的答案

96KB+16MB

參考答案

Total = 16,480 KB (~16.096 MB). Calculation: Direct = 12 \* 8KB = 96KB. Single Indirect = (8KB / 4B) \* 8KB = 2048 \* 8KB = 16,384KB.

Q14 問答題 (4分)

✓ 正確 4 / 4

List the four necessary conditions that must hold simultaneously for a system-wide deadlock to occur.

你的答案

- 1.mutual exclusion
- 2.hold and wait
- 3.no preemption
- 4.circular wait

參考答案

Mutual Exclusion; Hold and Wait; No Preemption; Circular Wait.

Q15 問答題 (3分)

✓ 正確 3 / 3

Describe the three requirements that a solution to the Critical Section problem must satisfy to be considered correct.

你的答案

- 1.mutual exculsion
- 2.progress
- 3.bounded wait

參考答案

Mutual Exclusion (only one in CS); Progress (selection of next process cannot be delayed indefinitely); Bounded Waiting (limit on how many times others can enter before a request is granted).

Q16 問答題 (3分)

✓ 正確 3 / 3

Explain the difference between 'Internal Fragmentation' and 'External Fragmentation'. Which one is typically associated with the Paging memory management scheme?

你的答案

Internal fragmentation:記憶體以固定大小block配置，若block大於需求，則block內剩餘的空間沒辦法分配給其他process>

External fragmentation:記憶體動態配置，空間被切成大大小小不連續的hole，即使總空間比需求大也可能找不到連續的block來滿足

paging memory management會有internal fragmentation的問題

參考答案

- a. Internal Fragmentation: Unused memory within an allocated fixed-size block.
- b. External Fragmentation: Total free memory exists but is not contiguous.
- c. Paging is associated with Internal Fragmentation (in the last page).

Q17 問答題 (3分)

✓ 正確 3 / 3

Compare the 'Many-to-One' and 'One-to-One' multithreading models. Which one allows better utilization of multi-core processors, and why?

你的答案

N:1 model:kernel只看得到一個process，看不到內部的threads。  
1:1 model:kernel看得到process內部的所有threads，scheduler可以分配每個threads到不同cores，可以parallel。  
所以1:1 model比較適合muti-core processors。

參考答案

- a. Many-to-One maps many user threads to one kernel thread.
- b. One-to-One maps each user thread to a kernel thread.
- c. One-to-One is better for multi-core because it allows multiple threads to run in parallel; Many-to-One can only run one thread at a time because the kernel only sees one entity.

Q18 簡答題 (2分)

✓ 正確 2 / 2

What is the 'Aging' technique in priority scheduling, and what specific problem does it solve?

你的答案

aging是指說隨著job在系統的時間越來越久，逐步提升它的優先權。主要是解決starvation(low priority job可能永遠無法完成工作)的問題

參考答案

- a. Aging gradually increases the priority of processes that wait in the system for a long time.
- b. It solves the problem of Starvation (indefinite blocking of low-priority processes).

Q19 簡答題 (2分)

✓ 正確 2 / 2

What is the primary purpose of the 'Translation Lookaside Buffer' (TLB) in a paging system?

你的答案

把常用的page先存在TLB，可以提升效率

參考答案

To speed up logical-to-physical address translation by caching frequently used page table entries, reducing the need for multiple memory accesses per reference.



**Q20 簡答題 (2分)**

✓ 正確 2 / 2

What is “local replacement” and “global replacement”?

你的答案

local replacement:執行中的process發生page fault 且 memory空間不夠要做page replacement時，只能從process自己被分配到的frames中挑選victim frames去做交換。global replacement:能從系統的physical memory裡面挑選victim frames做交換，所以可能挑到其他process的frames。

參考答案

- a. Local replacement: a process can only select a replacement frame from its own set of allocated frames.
- b. Global replacement: a process may select a replacement frame from the set of all memory frames.

**Q21 問答題 (3分)**

✓ 正確 3 / 3

What are the parameters of Multilevel Feedback Queue Scheduling? (List at least three)

你的答案

- 1.queue的數量
- 2.每個queue的排班
- 3.什麼情況下process要升級

參考答案

- a. number of queues
- b. scheduling algorithms for each queue
- c. methods used to determine when to upgrade a process
- d. methods used to determine when to demote a process
- e. methods used to determine which queue a process will enter when that process needs service

**Q22 簡答題 (2分)**

✓ 正確 2 / 2

What is the difference between a mode switch and a context switch?

你的答案

mode switch:CPU privileged level的改變，user mode <->kernel mode。context switch:把執行中的process A 的execution context存起來，載入另一個process B 的context，由A切換到B執行

參考答案

- a. Mode switch: change CPU execution mode from one privilege level to another, e.g., user → kernel via a trap or system call.
- b. Context switch: save one process' execution context & restore that of another process.

**Q23 簡答題 (2分)**

✓ 正確 2 / 2

A system has 32-bit addresses and 2 GB (231) main memory where each page is 1 MB (220). How many entries of a page table will contain?

你的答案

4096

參考答案

Each page is 1 MB (220), so it needs 20 bits as an offset. Thus a 32-bit address can have at most 12-bit (32-20=12) as the page number. Therefore a page table can have at most  $2^{12} = 4096$  entries.

**Q24 問答題 (2分)**

✓ 正確 2 / 2

Please briefly describe the Direct Memory Access (DMA)?

你的答案

是一種I/O技術，讓device可以在沒有CPU逐筆搬運data的情況下，直接在memory和device之間傳輸資料。CPU的工作是設定DMA參數並啟動DMA，之後DMA controller取得bus完成bulk transfer。完成之後會發送interrupt通知CPU，可以降低CPU在I/O的overhead，適合大量資料傳輸

參考答案

Direct Memory Access (DMA) is a technique used by computers to transfer data between devices without involving the CPU. DMA allows devices like hard drives, sound cards, and network cards to read or write data directly to or from the main memory, without the need for the CPU to intervene in every transfer, which would waste a significant amount of CPU time. DMA controllers can access the memory independently of the CPU and transfer large amounts of data at high speed, thereby reducing CPU usage and improving system performance.

**Q25 簡答題 (2分)**

✓ 正確 2 / 2

What is the race condition?

你的答案

在並行的系統中，兩個或兩個以上的process要存取某個共享資源，且至少有一方要寫入，由於操作non-atomic，缺乏正確的同步，導致結果會因為執行順序不同而不同

參考答案

A race condition occurs when two or more threads or processes access a shared resource such as a variable, file, or database record, and the final outcome of the program depends on the order in which the threads or processes execute.

Q26 單選題 (2分)

✓ 正確 2 / 2

A major problem with the base & limit approach to memory translation is:

- A. It requires the process to occupy contiguous memory locations.
- B. The translation process is time-consuming.
- C. The translation must be done for each memory reference.
- D. A process can easily access memory that belongs to another process.

你的選擇 正確

你的答案: A · 參考答案: A

Q27 單選題 (2分)

✓ 正確 2 / 2

Which state transition is not valid?

- A. Running → Ready
- B. Ready → Blocked
- C. Ready → Running
- D. Running → Blocked

你的選擇 正確

你的答案: B · 參考答案: B

Q28 單選題 (2分)

✓ 正確 2 / 2

Which disk scheduling algorithm is most vulnerable to starvation?

- A. SCAN.
- B. LOOK.
- C. Shortest Seek Time First (SSTF).
- D. First Come, First Served (FCFS)

你的選擇 正確

你的答案: C · 參考答案: C

Q29 單選題 (2分)

✓ 正確 2 / 2

A semaphore puts a thread to sleep:

- A. if it tries to decrement the semaphore's value below 0.
- B. if it increments the semaphore's value above 0.
- C. until another thread issues a notify on the semaphore.
- D. until the semaphore's value reaches a specific number.

你的選擇 正確

你的答案: A · 參考答案: A

**Q30 單選題** (2分)

✓ 正確 2 / 2

There are three processes on a system with the following performance requirements:

- . process A runs every 90 milliseconds, using 30 milliseconds of CPU time.
- . process B runs every 60 milliseconds, using 20 milliseconds of CPU time
- . process C runs every 500 milliseconds, using 10 milliseconds of CPU time

Use rate-monotonic analysis to assign priorities to the three processes. Assume that priorities range from 0 to 90, with 0 being the highest. Which of the following is a valid set of priority assignments?

- A. PA=40, PB=30, PC=100
- B. PA=40, PB=60, PC=20
- C. PA=70, PB=50, PC=60
- D. PA=50, PB=40, PC=90

你的選擇 正確

你的答案: D · 參考答案: D

**Q31 單選題** (2分)

✓ 正確 2 / 2

Which type of user thread and kernel thread mapping cannot take advantage of a multiprocessor?

- A. 1:1
- B. 1:M
- C. N:M
- D. N:1

你的選擇 正確

你的答案: D · 參考答案: D

**Q32 單選題** (2分)

✓ 正確 2 / 2

Which of the following statements about mount mechanism and the Virtual File System (VFS) is correct?

- A. The mount mechanism is used to divide one file into multiple blocks.
- B. The purpose of VFS is to allow a modern operating system to support only one file system.

The mount mechanism can combine multiple file systems into a single hierarchical namespace; VFS provides an abstraction layer that allows different file systems to be supported transparently.

你的選擇 正確

- D. VFS exists only in network devices, not in file systems.

你的答案: C · 參考答案: C

Q33 單選題 (2分)

✓ 正確 2 / 2

What is the main purpose of a timer interrupt in an operating system?

- A. To allow user programs to execute privileged instructions directly.
- B. To periodically return control to the OS so it can preempt processes, perform context switching, poll hardware status, and do bookkeeping.
- C. To copy data between disk and memory without CPU involvement.
- D. To make a blocked process skip the ready queue.

你的選擇 正確

你的答案: B · 參考答案: B

Q34 單選題 (2分)

✓ 正確 2 / 2

Which of the following statements about demand paging is correct?

- A. A valid bit indicates that the page is not in memory.
- B. An invalid bit can only indicate an out-of-bound reference.
- C. If there is an invalid reference, the OS will bring the missing page into memory; if the page is simply not in memory, the process will be aborted.
- D. A valid bit means the page is in memory; an invalid bit may indicate either an illegal reference or that the page is not in memory.

你的選擇 正確

你的答案: D · 參考答案: D

Q35 單選題 (2分)

✓ 正確 2 / 2

Thrashing occurs when:

- A. The sum of working sets of all processes exceeds available memory.
- B. The TLB hit ratio drops below 50%.
- C. Processes are switched too infrequently.
- D. I/O bursts overlap completely.

你的選擇 正確

你的答案: A · 參考答案: A

Q36 單選題 (2分)

✓ 正確 2 / 2

Which scheduling criterion measures the time from submission to completion?

- A. CPU utilization.
- B. Throughput.
- C. Turnaround time.
- D. Waiting time.

你的選擇 正確

你的答案: C · 參考答案: C

Q37 單選題 (2分)

✓ 正確 2 / 2

The File Allocation Table (FAT) in file systems is a variation of:

- A. Contiguous allocation.
- B. Combined indexing.
- C. Indexed allocation.
- D. Linked allocation.

你的選擇 正確

你的答案: D · 參考答案: D

Q38 單選題 (2分)

✓ 正確 2 / 2

Large page sizes in paging systems increase:

- A. Internal fragmentation.
- B. Page table size.
- C. External fragmentation.
- D. TLB efficiency.

你的選擇 正確

你的答案: A · 參考答案: A

Q39 單選題 (2分)

✓ 正確 2 / 2

Page fault frequency monitoring is used for:

- A. Optimizing TLB size.
- B. Reducing I/O bursts.
- C. Managing resident sets.
- D. Increasing priority inversion.

你的選擇 正確

你的答案: C · 參考答案: C

Q40 單選題 (2分)

✓ 正確 2 / 2

Which CPU scheduling algorithm is mathematically proven to provide the minimum average waiting time for a given set of processes?

- A. FCFS
- B. Round Robin
- C. Priority Scheduling

D. Shortest Job First (SJF)

你的選擇 正確

你的答案: D · 參考答案: D

Q41 單選題 (2分)

✓ 正確 2 / 2

Which deadlock prevention strategy works by requiring a process to request all its required resources at once?

- A. Mutual Exclusion
- B. Hold and Wait
- C. No Preemption
- D. Circular Wait

你的選擇 正確

你的答案: B · 參考答案: B

Q42 單選題 (2分)

✓ 正確 2 / 2

'Thrashing' is characterized by which of the following system behaviors?

- A. Very high paging activity and low CPU utilization.
- B. High CPU utilization.
- C. Rapid file deletions.
- D. Excessive disk capacity.

你的選擇 正確

你的答案: A · 參考答案: A

**Q43** 單選題 (2分)

✓ 正確 2 / 2

Peterson's Solution is a classic software-based solution for the Critical Section problem designed for:

- A. A single process.
- B. Exactly two processes.
- C. Any number of processes.
- D. Real-time systems.

你的選擇 正確

你的答案: B · 參考答案: B

**Q44** 單選題 (2分)

✓ 正確 2 / 2

In an Inverted Page Table, the number of entries in the table is equal to the number of:

- A. Virtual pages in logical memory.
- B. Threads in the process.
- C. Processes in the system.
- D. Physical frames in physical memory.

你的選擇 正確

你的答案: D · 參考答案: D